Calculation questions – 4.5 Forces

AQA GCSE Physics (8463) Higher · Paper 2 · 24 questions · 78 marks · answers at the back

Equations for this topic

Equation	Symbols and units	Exam
W = mg	W (N), m (kg), g (N/kg)	Learn it
W = Fs	W (J), F (N), s (m)	Learn it
F = ke	F (N), k (N/m), e (m)	Learn it
M = Fd (triple)	M (Nm), F (N), d (m)	Learn it
p = F/A (triple)	p (Pa), F (N), A (m²)	Learn it
s = vt	s (m), v (m/s), t (s)	Learn it
a = Δv/t	a (m/s²), Δv (m/s), t (s)	Learn it
F = ma	F (N), m (kg), a (m/s²)	Learn it
p = mv	p (kg m/s), m (kg), v (m/s)	Learn it
p = hρg (triple)	p (Pa), h (m), ρ (kg/m³), g (N/kg)	On sheet
$v^2 - u^2 = 2as$	v, u (m/s), a (m/s²), s (m)	On sheet
F = mΔv/Δt (triple, HT)	F (N), m (kg), Δv (m/s), Δt (s)	On sheet
Ee = ½ke²	Ee (J), k (N/m), e (m)	On sheet
stopping distance = thinking distance + braking distance		Rule
clockwise moments = anticlockwise moments (in balance)		Rule

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

Grade 4–5

1 Calculate the weight of a 65 kg person on Earth. (g = 9.8 N/kg) [2 marks]

Answer: _____ Unit: _____

2 A box is pushed 4.0 m with a force of 150 N. Calculate the work done. [2 marks]

Answer: _____ Unit: _____

3 A spring with spring constant 25 N/m is extended by 0.12 m. Calculate the force applied. [2 marks]

Answer: _____ Unit: _____

4 A car travels 450 m in 30 s. Calculate its average speed. [2 marks]

Answer: _____ Unit: _____

5 A resultant force of 2000 N acts on an 800 kg car. Calculate its acceleration. [2 marks]

Answer: _____ Unit: _____

6 (Triple) A 40 N force acts 0.25 m from a pivot. Calculate the moment. [2 marks]

Answer: _____ Unit: _____

Grade 6–7

7 A car accelerates from rest to 27 m/s in 9.0 s. Calculate its acceleration. [2 marks]

Answer: _____ Unit: _____

8 A train decelerates from 30 m/s to rest at 0.60 m/s². Calculate the time taken. [3 marks]

Answer: _____ Unit: _____

9 (Triple) A 600 N person stands on one foot of area 150 cm². Calculate the pressure on the ground in Pa. [3 marks]

Answer: _____ Unit: _____

10 A spring's length increases from 8.0 cm to 14 cm when a 4.2 N force is applied. Calculate the spring constant. [3 marks]

Answer: _____ Unit: _____

11 Calculate the momentum of a 0.16 kg ball travelling at 25 m/s. [2 marks]

Answer: _____ Unit: _____

12 A 75 kg astronaut weighs 120 N on the Moon. Calculate the gravitational field strength on the Moon. [3 marks]

Answer: _____ Unit: _____

13 (Triple) A child of weight 300 N sits 1.5 m from the pivot of a seesaw. How far from the pivot must a 450 N child sit on the other side to balance it? [3 marks]

Answer: _____ Unit: _____

14 A car starts from rest and accelerates at 2.5 m/s² over 80 m. Calculate its final speed. [3 marks]

Answer: _____ Unit: _____

Grade 8–9

15 A 1200 kg car travels at 20 m/s. The driver's reaction time is 0.70 s and the brakes decelerate the car at 6.5 m/s². Calculate (a) the stopping distance and (b) the braking force. [6 marks]

Answer: _____ Unit: _____

16 A 1.5 kg trolley moving at 4.0 m/s hits a stationary 2.5 kg trolley and they stick together. (a) Calculate their velocity after the collision. (b) Calculate the kinetic energy lost. [6 marks]

Answer: _____ Unit: _____

17 A 4.0 kg rifle fires a 0.020 kg bullet at 400 m/s. Calculate the recoil velocity of the rifle. [3 marks]

Answer: _____ Unit: _____

18 (Triple) A 70 kg driver travelling at 15 m/s is stopped by an airbag in 0.12 s. (a) Calculate the force on the driver. (b) Calculate the force if he hit the steering wheel and stopped in 0.020 s. (c) Explain how the airbag reduces injury. [6 marks]

Answer: _____ Unit: _____

19 (Triple) A diver is 25 m below the sea surface. Density of seawater = 1000 kg/m³. (a) Calculate the pressure due to the water. (b) Calculate the force from the water on a 0.040 m² mask window. [4 marks]

Answer: _____ Unit: _____

20 A cyclist accelerates uniformly from rest to 8.0 m/s in 10 s, then rides at 8.0 m/s for 20 s. Sketch the velocity–time graph and calculate (a) the total distance and (b) the mean speed. [5 marks]

Answer: _____ Unit: _____

21 An 80 kg skydiver experiences 500 N of air resistance. Calculate his acceleration. [3 marks]

Answer: _____ Unit: _____

22 A 60 kg person stands in a lift accelerating upwards at 1.5 m/s². Calculate the force from the floor on the person. [4 marks]

Answer: _____ Unit: _____

23 A spring (k = 800 N/m) is compressed by 5.0 cm and launches a 0.050 kg ball horizontally. Calculate the launch speed of the ball. [4 marks]

Answer: _____ Unit: _____

24 (Triple) A crowbar is 1.35 m long. A 200 N force is applied at one end, 1.2 m from the pivot. Calculate the force exerted on the load at the other end. [3 marks]

Answer: _____ Unit: _____

Answers – 4.5 Forces

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

Q	Working	Answer
1	$W = 65 \times 9.8$	**637 N**
2	$W = 150 \times 4.0$	**600 J**
3	$F = 25 \times 0.12$	**3.0 N**
4	$v = 450 \div 30$	**15 m/s**
5	$a = 2000 \div 800$	**2.5 m/s²**
6	$M = 40 \times 0.25$	**10 Nm**
7	$a = 27 \div 9.0$	**3.0 m/s²**
8	$t = 30 \div 0.60$	**50 s**
9	$A = 0.015$ m²; $p = 600 \div 0.015$	**40 000 Pa**
10	$e = 0.060$ m; $k = 4.2 \div 0.060$	**70 N/m**
11	$p = 0.16 \times 25$	**4.0 kg m/s**
12	$g = 120 \div 75$	**1.6 N/kg**
13	$300 \times 1.5 = 450 \times d$	**1.0 m**
14	$v^2 = 0 + 2 \times 2.5 \times 80 = 400$	**20 m/s**
15	thinking $= 20 \times 0.70 = 14$ m; braking: $0 - 20^2 = 2 \times (-6.5) \times s \rightarrow s = 30.8$ m; $F = 1200 \times 6.5$	**(a) 45 m (b) 7800 N**
16	(a) $1.5 \times 4.0 = 6.0$ kg m/s $= 4.0 \times v$ (b) before $\frac{1}{2} \times 1.5 \times 4.0^2 = 12$ J; after $\frac{1}{2} \times 4.0 \times 1.5^2 = 4.5$ J	**(a) 1.5 m/s (b) 7.5 J**
17	total momentum $= 0$: $0.020 \times 400 = 4.0 \times v$	**2.0 m/s backwards**
18	$\Delta p = 70 \times 15 = 1050$ kg m/s; (a) $1050 \div 0.12$ (b) $1050 \div 0.020$ (c) longer stopping time → smaller rate of change of momentum → smaller force	**(a) 8750 N (b) 52 500 N**
19	(a) $p = 25 \times 1000 \times 9.8$ (b) $F = 245\,000 \times 0.040$	**(a) 245 000 Pa (b) 9800 N**
20	area: $\frac{1}{2} \times 10 \times 8.0 = 40$ m, plus $20 \times 8.0 = 160$ m; mean $= 200 \div 30$	**(a) 200 m (b) 6.7 m/s**
21	weight $= 784$ N; resultant $= 784 - 500 = 284$ N; $a = 284 \div 80$	**3.6 m/s² downwards**
22	$R - mg = ma \rightarrow R = 60 \times 9.8 + 60 \times 1.5$	**678 N**
23	$Ee = \frac{1}{2} \times 800 \times 0.050^2 = 1.0$ J; $1.0 = \frac{1}{2} \times 0.050 \times v^2 \rightarrow v^2 = 40$	**6.3 m/s**
24	load distance $= 1.35 - 1.2 = 0.15$ m; $200 \times 1.2 = F \times 0.15$	**1600 N**

Common mistakes. Using length instead of extension (Q10); forgetting thinking distance (Q15); using weight instead of resultant force in F = ma (Q21, Q22); forgetting momentum is a vector (Q17).